

Mohammad Ratul Mahjabin

mohammadratul@usf.edu | (813) 3605597 | www.ratul-mahjabin.github.io

Research Interests

Computational Social Science, Social Media Mining, Machine Learning, Data Science, Network Science, Computer Vision, Human-Computer Interaction, Natural Language Processing, Statistical Modeling

Education

University of South Florida, Tampa, FL

Ph.D. in Computer Science and Engineering

Jan '24 - Present

Research Areas: Computational Social Science, Human-Computer Interaction, NLP

Advisor: Dr. Raiyan Abdul Baten - [Computational Social Science and AI Lab](#)

Islamic University of Technology, Dhaka, Bangladesh

B.Sc. in Software Engineering

Jan '19 - May '23

Thesis: [An Efficient Deep Learning-based approach for Recognizing Agricultural Pests in the Wild](#)

Publications

Recent preprints under review

- P1 **M. R. Mahjabin** and R. A. Baten, [The Wisdom of Intellectually Humble Networks](#), [arxiv:2502.02015](#), 2025

Journal Papers

- J3 Mustafina SN, Islam MN, **Mahjabin MR**, Mannan MM, Islam MM. [Identifying key indicators to develop a novel mobile application for early screening of postpartum depression in developing countries](#). BMC Health Services Research. 2025 Dec;25(1):1-20.
- J2 Rafi TH, **Mahjabin R**, Ghosh E, Ko YW, Lee JG. [Domain generalization for semantic segmentation: a survey](#). Artificial Intelligence Review. 2024 Aug 12;57(9):247.
- J1 Islam MN, Jahangir R, Mohim NS, Wasif-UI-Islam M, Ashraf A, Khan NI, **Mahjabin MR**, Miah AS, Shin J. [A multilingual handwriting learning system for visually impaired people](#). IEEE Access. 2024 Jan 15.

Conference Papers

- C2 Hema NE, Zeest NS, **Mahjabin MR**, Islam MN. [An Ensemble-based ML Model to Predict Cardiac Disease](#). In 2024 International Conference on Inventive Computation Technologies (ICICT) 2024 Apr 24 (pp. 214-219). IEEE
- C1 Rafi MH, **Mahjabin MR**, Rahman MS, Kabir MH, Ahmed S. [A Critical Analysis of Deep Learning Applications in Crop Pest Classification: Promising Pathways and Limitations](#). In 2023 26th International Conference on Computer and Information Technology (ICCIT) 2023 Dec 13 (pp. 1-6). IEEE.

Professional Experience

University of South Florida, Tampa, FL

Graduate Research and Teaching Assistant, Dept. of Computer Science and Engineering Jan '24 - Current

Military Institute of Science and Technology (MIST), Dhaka, Bangladesh

Research Assistant, Dept. of Computer Science and Engineering Aug '23 - Dec '23

Samsung R&D Institute, Dhaka, Bangladesh

Software Engineer Intern May '22 - Sept '22

Selected Projects

PPD Coach

Mobile Application

An android application for Screening Postpartum Depression in the context of Bangladesh 2023

- It screens postpartum depression using story-based questions and visual aids.
- Developed with: [Flutter](#), [Firebase](#)

ScrapeText

Notebook

Extracting articles from various online sources using Python [\[Github\]](#) 2023

- Text scraping tool designed to extract and perform textual analysis calculating different metrics
- Developed with: [Python](#), [Pandas](#), [NLTK](#), [Regex](#)

FoodScan101

WebApp

Enables users to detect and identify different types of food from a photo [\[Github\]](#) 2023

- A CNN(EfficientNetV2) was fine-tuned through transfer learning on a [dataset](#)
- Developed with: [Python](#), [Flask](#), [CNN](#)

HaateKhorl

WebApp

ML-powered web app for early childhood education [\[Github\]](#) 2021

- This project utilizes ML techniques to provide early education to children in an engaging and interactive way [\[Video Demo\]](#)
- Developed with: [Python](#), [OpenCV](#), [YOLO](#), [React](#), [Nodes.js](#)

Covid-CXRNet

Notebook

Detecting Covid19 with Chest X-ray using CNN [\[Github\]](#) 2021

- This project implements a CNN model(ResNet18) to accurately detect COVID-19 with an accuracy of 99.98%
- Developed with: [Python](#), [Pytorch](#)

Skills

Programming Python, R, C/C++, Javascript, Java, HTML, CSS

Libraries PyTorch, TensorFlow, NetworkX, NLTK, Pandas, Scikit-Learn, Matplotlib, Numpy

Web & DB Flask, Django, React, MongoDB, Firebase, REST API

Tool Docker, Heroku, Git, LaTeX